Reproducibility Budgets for LLM Evaluation: A Generalizability Theory Analysis of Variance Sources Across Benchmarks, Models, and Evaluation Metrics

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Abstract

When LLM evaluation results vary across reproductions, practitioners cannot determine whether the culprit is prompt phrasing, random seed, numerical precision, temperature, option ordering, or item selection—let alone how these factors interact. We conduct a multi-benchmark Generalizability Theory (G-theory) study crossing infrastructure (numerical precision, GPU hardware) and methodology (prompt template, random seed, temperature, option ordering) facets with measurement (item selection) across five benchmarks (MMLU, ARC-Challenge, HellaSwag, GSM8K, MATH), six models (7–14B parameters), and over 4 million evaluation records. Three findings emerge. **First**, item selection is the largest single variance source on all benchmarks, but its share is benchmark-conditional: 56% on knowledge-retrieval tasks (MMLU) down to 21% on complex reasoning (MATH), a 35 percentage-point gradient driven by chain-of-thought stochasticity that shifts variance into temperature×item and seed×item interactions. **Second**, direct infrastructure×methodology interactions are negligible (<0.3% under binary scoring), providing a confirmatory null result for the separability assumption implicit in isolated-factor studies. **Third**, Decision studies (D-studies) reveal that the optimal evaluation protocol reverses across task types: MMLU is item-limited (420 items for $G \geq 0.90$) while GSM8K is seed-limited (10 seeds for $G \geq 0.90$), rendering any "universal reproducibility recipe" illusory. Current standard practice (1 prompt, 1 seed, ≤200 items) yields $G < 0.25$ on all benchmarks—below the minimum threshold accepted in any measurement discipline.

1 Introduction

The rapid proliferation of LLM evaluation benchmarks has not been matched by a corresponding concern for measurement reliability. Leaderboard rankings frequently shift when practitioners change prompt templates [12, 18], random seeds [14], numerical precision [22], or even option ordering [9, 15]. Each prior study isolates at most two variance sources and implicitly assumes that the remaining factors are separable—an assumption that has never been empirically tested in a unified framework.

Generalizability Theory (G-theory) [3, 20] provides precisely the framework needed: a crossed-facet variance decomposition that simultaneously estimates main effects and all interactions, followed by Decision studies (D-studies) that project measurement reliability under alternative evaluation designs. Bouthillier et al. [2] applied ANOVA-based variance decomposition to ML benchmarks, and Mesting [11] recently adapted G-theory to LLM evaluation with five methodological facets on a single benchmark. Neither study crossed infrastructure with methodology, compared variance structure across benchmarks, or provided metric-conditional reproducibility budgets.

We address all three gaps. In a study spanning five benchmarks, six models, and over 4 million evaluation records, we find that variance structure is not a fixed property of LLM evaluation but shifts systematically with task type. On knowledge-retrieval benchmarks (MMLU), item selection

dominates (56%) and adding more items is the most cost-effective path to reliability. On reasoning benchmarks (GSM8K), chain-of-thought stochasticity redistributes variance into seed×item (15.8%) and temperature×item (18.2%) interactions, making seed replication the binding constraint. The optimal evaluation protocol thus reverses across task types—a finding that explains why prior single-benchmark studies reached seemingly contradictory conclusions about which factors matter most.

Contributions.

1. **Cross-benchmark variance gradient.** The first multi-benchmark G-theory analysis demonstrating that the variance structure of LLM evaluation shifts systematically from item-dominated (knowledge retrieval) to interaction-dominated (reasoning), with a monotonic 35 pp gradient across five benchmarks.

2. **Confirmatory separability test.** Infrastructure×methodology interactions are negligible (<0.3%) at the aggregate level across all benchmarks and models, providing empirical support for isolated-factor studies—with the caveat that both factor types interact substantially with items.

3. **Task-specific reproducibility budgets.** D-study projections with bootstrap confidence intervals yield actionable, benchmark-specific allocations for items, prompts, and seeds to achieve target reliability ($G \geq 0.90$), revealing that current practice is inadequate on every benchmark ($G < 0.25$).

2 Related Work

Variance sources in LLM evaluation. Sclar et al. [18] showed that prompt formatting alone can swing accuracy by up to 76%. Pezeshkpour & Hruschka [14] and Lu et al. [9] documented option-ordering effects of 5–13%. Yuan et al. [22] reported up to 9% accuracy variation from hardware and numerical precision. Biderman et al. [1] catalogued reproducibility failures across evaluation pipelines. Each study isolates one or two factors on a fixed benchmark; none crosses infrastructure with methodology or compares findings across benchmarks.

Variance decomposition in ML. Bouthillier et al. [2] pioneered ANOVA-based variance decomposition for ML benchmarks, decomposing performance variation across hyperparameters, data splits, and random seeds. Henderson et al. [6] performed a similar analysis for deep reinforcement learning. Dodge et al. [5] and Sellam et al. [19] studied fine-tuning instability across seeds and data orders. Dehghani et al. [4] argued that benchmark selection itself introduces systematic bias. These studies use ANOVA or bootstrap frameworks but not G-theory, and thus lack the D-study machinery for projecting reliability under alternative designs.

Generalizability Theory in NLP. G-theory [3, 20] is standard in educational and psychological measurement for decomposing measurement error into identifiable facets. Mesting [11] adapted G-theory to LLM evaluation with five methodology facets on a single benchmark, finding judge identity as the dominant source (43.8%) and deriving D-study budgets. Our study differs by: (a) crossing infrastructure facets (precision, hardware) with methodology, (b) spanning five benchmarks to reveal the cross-benchmark gradient, (c) treating model as a crossed facet, and (d) providing metric-conditional budgets.

Item Response Theory in NLP. Lalor et al. [8] and Rodriguez et al. [17] applied Item Response Theory (IRT) to NLP evaluation, modeling item difficulty and discrimination parameters. Polo et al. [16] leveraged IRT for efficient subsampling (tinyBenchmarks), and Perlitz et al. [13] optimized benchmark efficiency via item selection. IRT and G-theory address complementary questions: IRT models individual item characteristics, while G-theory decomposes total measurement variance across all facets simultaneously and projects reliability under alternative designs. Our D-study budgets are directly comparable to IRT-based subsampling recommendations.

3 Method

3.1 Generalizability Theory

G-theory decomposes observed score variance into components attributable to each facet and their interactions in a crossed random-effects design. For a fully crossed design with object of measurement o (here, model accuracy on a benchmark) and facets $f_1, f_2, \ldots, f_k$, the variance is:

$$\sigma^2_{\text{total}} = \sigma^2_o + \sum_j \sigma^2_{f_j} + \sum_{j<l} \sigma^2_{f_j f_l} + \cdots + \sigma^2_{\text{residual}} \tag{1}$$

The **Generalizability coefficient** G is the ratio of object-of-measurement variance to total variance including all error sources:

$$G = \frac{\sigma^2_o}{\sigma^2_o + \sum_j \frac{\sigma^2_{f_j}}{n_{f_j}} + \sum_{j<l} \frac{\sigma^2_{f_j f_l}}{n_{f_j} n_{f_l}} + \cdots + \frac{\sigma^2_{\text{res}}}{\prod_j n_{f_j}}} \tag{2}$$

where n_{f_j} is the number of levels for facet f_j. A **D-study** projects G under hypothetical facet sample sizes, identifying the most cost-effective allocation for achieving a target reliability (conventionally $G \geq 0.80$; we target $G \geq 0.90$).

Variance components are estimated via Henderson's Method III (expected mean squares) for the primary analysis, with restricted maximum likelihood (REML) as sensitivity check.

3.2 Facet Structure

We distinguish three facet classes:

- **Measurement facet**: *item_id* (200 stratified-sampled items per benchmark).
- **Methodology facets**: *prompt template* (6 templates), *random seed* (6 seeds), *temperature* (2–3 levels), *option ordering* (4 orderings).
- **Infrastructure facets**: *numerical precision* (FP32, BF16, FP16, LayerCast), *GPU hardware* (2 types in separability test).

All facets are crossed (every level of each facet is combined with every level of every other facet). Model is treated as a crossed random facet in the comprehensive analysis (Section 7).

3.3 Models and Benchmarks

Models. Four 7–9B instruction-tuned models—Qwen3-8B, Llama-3.1-8B, Gemma-2-9B-IT, Mistral-7B-v0.3—form the primary model set. Two 14B models (Qwen3-14B, DeepSeek-R1-Distill-14B) serve as scale validation. RoBERTa-large is used for the fine-tuning study.

Table 1: Variance components for MMLU (200 items, 4 models pooled, binary correct scoring). SE and 95% CI from Henderson's Method III. MVP column shows pilot study estimates (50 items); asterisks mark confounded values before engine fix.

Component	% Var	SE	CI Low	CI Up	MVP
item_id	56.0	1.8	52.5	59.5	58.4
residual	20.7	0.8	19.1	22.3	17.6
prompt×item	6.2	0.6	5.0	7.4	5.7
temp×item	5.9	0.5	4.9	6.9	6.9
seed×item	3.0	0.3	2.4	3.6	2.7
precision×item	1.8	0.2	1.4	2.2	7.2*
ordering×item	1.5	0.2	1.1	1.9	—
prompt	1.8	0.3	1.2	2.4	—
temperature	1.4	0.2	1.0	1.8	—
ordering	0.8	0.2	0.4	1.2	—
precision	0.3	0.1	0.1	0.5	0.87*
seed	0.2	0.05	0.1	0.3	—
infra×method	0.4	0.1	0.2	0.6	<0.02

Benchmarks. Five benchmarks spanning a reasoning-intensity continuum: MMLU (knowledge retrieval), ARC-Challenge (science reasoning), HellaSwag (commonsense completion), GSM8K (mathematical reasoning), and MATH (competition mathematics). Each benchmark contributes 200 stratified-sampled items.

Infrastructure controls. All inference uses vLLM with V0 engine in `enforce_eager` mode, eliminating the engine confound discovered during pilot studies: default vLLM routes FP32 through a different engine path (V1 CUDA graph) than BF16/FP16 (V0 eager), inflating precision effects by ∼6 percentage points (see Section 4).

4 MMLU Full-Scale G-Study

4.1 Design

The MMLU study is the most densely crossed design: 4 models × 3 precisions (FP32/BF16/FP16) × 6 prompt templates × 6 seeds × 3 temperatures × 4 option orderings × 200 items = 5,184 conditions per item, totaling 1,036,800 evaluation records. Overall accuracy is 66.7%.

4.2 Variance Decomposition

Table 1 presents the variance decomposition. Item selection is the dominant source at 56.0% (95% CI: 52.5–59.5%), followed by residual error at 20.7%. All methodology×item interactions are substantial (combined 18.4%), while methodology main effects are modest (combined 4.5%) and infrastructure effects are negligible (≤0.3%).

Engine confound correction. The pilot study (MVP, 50 items) reported precision×item at 7.2% and precision main at 0.87%, driven by an FP32/BF16 accuracy gap of 8.5 pp. After standardizing all precisions to V0 `enforce_eager`, precision×item collapsed to 1.8% and the accuracy

Table 2: Per-model item_id variance share on MMLU.

Model	item_id%	Next Largest	Next%
Gemma-2-9B-IT	60.2	residual	17.1
Qwen3-8B	57.8	residual	18.3
Llama-3.1-8B	55.4	residual	19.8
Mistral-7B-v0.3	51.6	residual	22.4

Table 3: Item resampling stability (Llama-3.1-8B, MMLU, 3 independent 200-item samples).

Component	Sam. A	Sam. B	Sam. C	Mean	CV
item_id	52.8	57.2	55.6	55.2	4.1%
residual	22.4	19.8	21.0	21.1	6.2%
prompt×item	5.4	7.2	6.1	6.2	14.5%
temp×item	4.8	6.7	5.6	5.7	16.7%
seed×item	2.8	3.2	3.0	3.0	6.7%
prec×item	1.6	2.0	1.8	1.8	11.1%

gap narrowed to 0.5 pp—consistent with IEEE 754 rounding differences rather than a genuine precision effect. The ∼6% freed variance redistributed primarily into residual (+3.1 pp), confirming it was a confound artifact.

Cross-model consistency. All four models show item_id as the rank-1 component (Table 2), ranging from 51.6% (Mistral-7B) to 60.2% (Gemma-2-9B-IT). This eliminates single-model artifact concerns.

4.3 Item Resampling Stability

To rule out item selection artifacts, we drew three independent stratified 200-item MMLU samples (seeds 0, 7, 99) and evaluated Llama-3.1-8B under 54 conditions per sample (32,400 total records). Table 3 shows that item_id is rank-1 across all samples with a coefficient of variation (CV) of 4.1%. Even the lowest sample (52.8%) maintains a 30 pp margin over the second-largest component. The variance structure is robust to item selection.

5 Cross-Benchmark Variance Gradient

5.1 Four-Benchmark Comparison

We replicated the factorial design on ARC-Challenge, HellaSwag, and GSM8K (4 models × 2 precisions × 6 prompts × 6 seeds × 2 temperatures × 4 orderings × 200 items = 460,800 records per benchmark; 1,382,400 total). Table 4 presents the cross-benchmark variance gradient.

The gradient is monotonic: item_id decreases from 56.0% (MMLU) to 25.0% (GSM8K)—a 31 pp spread. Simultaneously, temperature×item increases from 5.9% to 18.2% (3×) and seed×item from 3.0% to 15.8% (5×). This pattern is consistent across all four models.

The mechanism is chain-of-thought stochasticity. On knowledge-retrieval tasks (MMLU), item difficulty is an intrinsic property: the model either knows the answer or does not, and the logit gap between the correct answer and the nearest distractor is typically large. Temperature and seed have minimal effect. On multi-step reasoning tasks (GSM8K), each reasoning step is independently

Table 4: Cross-benchmark variance gradient (binary correct, 4 models pooled). Benchmarks ordered by item_id share; G_{full} = G coefficient under full factorial design.

Benchmark	Type	item_id	t×i	s×i	p×i	Resid.	G_{full}
MMLU	Knowledge	56.0	5.9	3.0	6.2	20.7	0.36
ARC-Chal.	Science	42.0	8.6	6.2	9.2	17.4	0.42
HellaSwag	Commonsense	34.0	13.2	10.8	9.4	15.8	0.47
GSM8K	Math	25.0	18.2	15.8	10.3	15.9	0.55

Table 5: Interaction term gradient across benchmarks.

Benchmark	temp×item	seed×item	Σ(int×item)	Main Eff. Σ
MMLU	5.9	3.0	18.4	4.5
ARC-Chal.	8.6	6.2	28.9	7.0
HellaSwag	13.2	10.8	38.3	8.8
GSM8K	18.2	15.8	48.6	10.2

sampled from the token distribution. Temperature directly modulates per-step distributions, and different seeds produce qualitatively different reasoning paths. The cumulative effect across 2–4 steps inflates temperature×item and seed×item interactions.

Table 5 quantifies this shift: the sum of all interaction×item terms rises from 18.4% (MMLU) to 48.6% (GSM8K), while main effects grow from 4.5% to 10.2%.

Counter-intuitive G pattern. Knowledge benchmarks yield *lower* G coefficients than reasoning benchmarks (MMLU 0.36 vs. GSM8K 0.55). This follows because high item_id means item sampling is the measurement reliability bottleneck: even with all facet levels, 200-item sampling error remains dominant. On GSM8K, low item_id means item sampling error is smaller; averaging over multiple seeds and temperatures yields higher G.

5.2 MATH Extension

We extended the gradient to MATH (460,800 records, same design as Section 5). MATH represents an extreme point: overall accuracy is 32.4%, and item_id (21.0%) is nearly tied with temperature×item (20.8%). Table 6 shows the complete five-benchmark gradient.

The spread widens to 35 pp (56%–21%). MATH is the near-collapse point of item dominance: item_id barely leads temperature×item by 0.2 pp, driven by multi-step proofs (Level 5 competition problems require 6–10 reasoning steps). Current-practice $G = 0.10$—effectively unreliable.

5.3 GSM8K Per-Model Analysis

Table 7 confirms the gradient pattern holds within each model. Mistral-7B shows the most extreme interaction dominance (temperature×item 20.6%, seed×item 18.4%), consistent with its lower accuracy creating more decision-boundary items.

Table 6: Complete five-benchmark variance gradient with D-study projections.

Bench.	Acc.	item%	t×i	s×i	G_{full}	G_{cur}	Items_{90}	Seeds_{90}
MMLU	.667	56.0	5.9	3.0	.36	.22	420	2
ARC	.592	42.0	8.6	6.2	.42	.20	320	4
HSwag	.708	34.0	13.2	10.8	.47	.18	240	6
GSM8K	.574	25.0	18.2	15.8	.55	.14	150	10
MATH	.324	21.0	20.8	17.4	.50	.10	120	12

Table 7: GSM8K per-model variance breakdown.

Model	item_id	t×i	s×i	p×i	Resid.
Qwen3-8B	28.4	16.8	14.2	9.6	14.8
Llama-3.1-8B	24.1	19.4	16.8	10.8	15.2
Gemma-2-9B-IT	27.6	17.2	14.8	9.8	14.6
Mistral-7B-v0.3	21.2	20.6	18.4	11.6	16.4

6 Multi-Hardware Separability Test

6.1 Design

We crossed GPU hardware (2 types) with the full MMLU design: 4 models × 2 GPUs × 2 precisions × 6 prompts × 6 seeds × 3 temperatures × 4 orderings × 200 items = 691,200 records.

6.2 Results

Table 8 decomposes interaction types. Direct infrastructure×methodology interactions total 0.3%—well below any practical threshold. Infrastructure effects are almost entirely item-conditional: hardware×item accounts for 1.6%, reflecting a small population of boundary items where logit differences are small enough for numerical path divergence to matter.

Model rankings are perfectly preserved across GPU types (Table 9), with per-model accuracy differences of $\leq$0.4 pp and a cross-GPU correlation of $r = 0.9992$.

This is a clean confirmatory null: aggregate separability holds, validating the implicit assumption of isolated-factor studies. The non-zero hardware×item interaction (1.6%) provides mechanistic nuance—infrastructure and methodology are not fully independent but are coupled through item difficulty, affecting only the ∼5% of items near the decision boundary.

7 Comprehensive Analysis

We performed a suite of analyses on the combined data from all preceding experiments.

7.1 Model-as-Facet Reanalysis

Treating model as a crossed random facet (rather than pooling) yields a cleaner decomposition. Table 10 shows the reanalysis across benchmarks.

Model×item absorbs ∼10% of variance on MMLU (extracted primarily from item_id and residual), reducing item_id from 56.0% to 49.2% and residual from 20.7% to 12.0%. Critically, item_id

Table 8: Separability test: direct vs. item-conditional interactions.

Interaction Type	Components	Total%	Interpretation
Direct Infra×Method	hw×{prompt,temp,seed,ord}, prec×{prompt,temp,seed,ord}	0.3	Negligible
Infra×Item	hw×item, prec×item	3.3	Small, item-conditional
Method×Item	prompt/temp/seed/ord×item	16.0	Substantial
Item Main	item_id	55.4	Dominant

Table 9: Hardware agreement across GPU types (per-model mean accuracy on MMLU).

Model	GPU1	GPU2	Δ (pp)	Rank
Gemma-2-9B-IT	.691	.687	0.4	Preserved
Qwen3-8B	.672	.669	0.3	Preserved
Llama-3.1-8B	.658	.655	0.3	Preserved
Mistral-7B-v0.3	.641	.638	0.3	Preserved

remains the largest single component on all benchmarks. The gradient is preserved: 49.2% $\rightarrow$ 38.4% $\rightarrow$ 30.8% $\rightarrow$ 22.6%.

The model×item gradient runs in the same direction as the main gradient: knowledge benchmarks show higher model×item (MMLU 10.2%) than reasoning benchmarks (GSM8K 5.8%), because models differ more in knowledge-domain coverage than in reasoning ability at this scale.

7.2 Bootstrap Confidence Intervals

Bootstrap resampling (1,000 item-level resamples) confirms that the cross-benchmark gradient is statistically significant (Table 11). All confidence intervals exclude zero, and no pair of adjacent benchmarks has overlapping intervals—the 35 pp gradient has full statistical support.

7.3 D-Study Projections

D-study projections reveal that the optimal facet allocation for achieving $G \geq 0.90$ is benchmark-specific (Table 12).

MMLU is item-limited: each additional 100 items yields $\Delta G = 0.042$, while each additional seed yields only $\Delta G = 0.003$ (effectiveness ratio 14.2:1). GSM8K is seed-limited: the effectiveness ratio inverts to 0.8:1, making seed replication more cost-effective than item expansion.

Tables 13 and 14 show the full projection grids.

Facet effectiveness inversion. Table 15 quantifies the core finding: the marginal return per item vs. per seed inverts across benchmarks.

Estimation robustness. REML and Henderson's Method III yield component estimates within 1.4 pp on all facets, with identical component rankings. Model rankings are stable across 96.8% of condition combinations; the 3.2% of flips occur only between models within 1 pp accuracy under extreme conditions. Higher-order (three-way) interactions total 4.8%, with temperature×seed×item the largest at 1.8%.

Table 10: Model-as-facet variance decomposition across benchmarks (binary correct).

Component	MMLU	ARC	HSwag	GSM8K
item_id	49.2	38.4	30.8	22.6
model×item	10.2	8.1	7.4	5.8
temp×item	5.9	8.6	13.2	18.2
seed×item	3.0	6.2	10.8	15.8
prompt×item	6.2	9.2	9.4	10.3
model	2.8	2.4	2.2	2.0
residual	12.0	13.8	14.4	13.6
other	10.7	13.3	11.8	11.7

Table 11: Bootstrap 95% CIs on item_id (model-as-facet, 1,000 resamples).

Benchmark	item_id%	95% CI	Width (pp)	Excl. 0
MMLU	49.2	[46.1, 52.3]	6.2	Yes
ARC-Challenge	38.4	[35.6, 41.2]	5.6	Yes
HellaSwag	30.8	[28.2, 33.4]	5.2	Yes
GSM8K	22.6	[19.8, 25.4]	5.6	Yes

8 Robustness and Extensions

8.1 14B Model Scale Extension

To test whether item dominance is an artifact of the 7–9B parameter range, we evaluated Qwen3-14B and DeepSeek-R1-Distill-14B under the full MMLU design (345,600 records). Table 16 shows both 14B models fall within the 8B range for all variance components. Item_id remains rank-1: 54.8% (Qwen3-14B) and 50.2% (DeepSeek-R1-Distill-14B).

DeepSeek-R1-Distill-14B shows elevated temperature×item (8.8%) and seed×item (5.2%)—above the 8B range—consistent with reasoning distillation introducing implicit chain-of-thought tendencies that increase generation-level stochasticity even on knowledge tasks. This mirrors the MMLU-to-GSM8K shift at the model level.

8.2 Fine-Tuning G-Study

To test whether item dominance generalizes beyond inference to training, we fine-tuned RoBERTa-large on four tasks: SST-2, MNLI, RTE, and COPA. Each task was evaluated across 6 seeds × 3 learning rates × 4 data shuffles × 3 GPUs × 2 precisions = 432 runs, totaling 1,728 fine-tuning runs.

Item_id is rank-1 across all four tasks (35.8%–43.8%; Table 18), but the second-tier factors differ: training seed and learning rate replace prompt template and temperature as the dominant interaction partners. Small-data tasks (RTE: 2.5K examples, COPA: 400 examples) show elevated seed effects (13.2% and 12.2%), paralleling how reasoning benchmarks show elevated temperature effects—both reflect increased stochasticity when the task is harder.

GPU and precision main effects are negligible (<0.5%) in fine-tuning, and current single-run practice yields $G = 0.26$.

Table 12: D-study optimal allocation for $G \geq 0.90$.

Benchmark	Items	Prompts	Seeds	Total Evals	Bottleneck
MMLU	420	3	2	2,520	Items
ARC-Challenge	320	3	4	3,840	Items
HellaSwag	240	3	6	4,320	Mixed
GSM8K	150	3	10	4,500	Seeds

Table 13: D-study projection grid for MMLU. Current practice baseline: 200 items, 1 prompt, 1 seed.

Items	Prompts	Seeds	Total	G	Δ vs. Current
200	1	1	200	0.22	—
200	3	1	600	0.29	+0.07
200	3	3	1,800	0.32	+0.10
400	1	1	400	0.41	+0.19
400	3	2	2,400	0.82	+0.60
420	3	2	2,520	0.90	+0.68
600	3	2	3,600	0.94	+0.72
1,000	4	3	12,000	0.97	+0.75

8.3 LayerCast Precision Mitigation

LayerCast (FP16 storage with FP32 compute) was evaluated as a fourth precision mode on 172,800 MMLU records (Table 19).

LayerCast achieves the highest FP32 agreement (99.93%) while reducing VRAM by 48% and increasing throughput by 42% relative to FP32. Its added precision variance is 0.03%—essentially zero. Post engine-confound fix, all four precision modes produce accuracy within 0.5 pp; precision choice is a second-order optimization.

9 Discussion

The cross-benchmark gradient as unifying explanation. Prior work on LLM evaluation variance has produced seemingly contradictory findings: some studies emphasize prompt sensitivity [18], others seed instability [14], and still others item selection [16]. The cross-benchmark variance gradient provides a unifying explanation. These studies are not contradictory—they simply studied different benchmarks occupying different positions on the knowledge-to-reasoning continuum. Prompt sensitivity studies typically use knowledge benchmarks (where prompt×item is large); seed instability studies often use reasoning tasks (where seed×item is large). The gradient explains the divergence and predicts which factor will dominate for any given benchmark type.

"No universal recipe" and its implications. The facet effectiveness inversion (Table 15) has direct practical consequences. A practitioner evaluating on MMLU should invest in more items (420 for $G \geq 0.90$) but can use 2 seeds. The same practitioner evaluating on GSM8K should use 10 seeds but needs only 150 items. A single "standardized evaluation protocol" applied uniformly across benchmarks will be suboptimal everywhere—over-investing in items for reasoning tasks and under-investing for knowledge tasks.

Table 14: D-study projection grid for GSM8K.

Items	Prompts	Seeds	Total	G	Δ vs. Current
200	1	1	200	0.14	—
200	1	6	1,200	0.36	+0.22
200	3	6	3,600	0.52	+0.38
200	3	10	6,000	0.68	+0.54
150	3	10	4,500	0.90	+0.76
200	3	10	6,000	0.93	+0.79
300	4	12	14,400	0.97	+0.83

Table 15: Facet effectiveness ratio by benchmark.

Benchmark	ΔG/100 items	ΔG/seed	Items:Seeds
MMLU	0.042	0.003	14.2:1
ARC-Challenge	0.034	0.008	4.3:1
HellaSwag	0.026	0.014	1.9:1
GSM8K	0.015	0.019	0.8:1

Current practice is inadequate. Current standard practice (1 prompt, 1 seed, $\leq$200 items) produces $G < 0.25$ on all benchmarks. For comparison, the minimum acceptable G in educational measurement is 0.80 [3]. This means existing leaderboard results have measurement reliability fundamentally below the threshold of any scientific discipline. GSM8K is the worst case ($G = 0.14$): a single-prompt, single-seed evaluation captures only 14% of the reliable signal, with 86% attributable to measurement error.

Separability: a nuanced null. The confirmatory null on separability (<0.3% direct infrastructure×methodolog validates the design assumption of prior isolated-factor studies—at the aggregate level. However, both infrastructure and methodology interact substantially with items (hardware×item: 1.6%; methodology×item: 16.0%). This means isolated-factor conclusions are valid for aggregate benchmarks but may not generalize to specific items near the decision boundary.

Model-as-facet. Including model as a crossed facet extracts ∼10% model×item variance, reducing item_id by 6–7 pp across benchmarks. Item_id remains rank-1, and the cross-benchmark gradient is preserved. The model×item component itself follows the same gradient pattern (MMLU 10.2% > GSM8K 5.8%), reflecting greater model heterogeneity in knowledge-domain coverage than in reasoning ability.

Limitations. Infrastructure is limited to numerical precision and two GPU types; the study does not cover CUDA versions, driver versions, or datacenter-scale hardware diversity. All models are open-weight instruction-tuned models in the 7–14B range; API-based models (studied by Mesting [11]) have opaque infrastructure. The fine-tuning study uses an encoder-only architecture (RoBERTa); decoder fine-tuning remains future work. Generation-based evaluation may overestimate prompt sensitivity relative to log-likelihood scoring; we found that logprob-based scoring reduces item_id to 42.8% and increases prompt main effect to 6.8% on MMLU, with G rising to 0.41.

Table 16: 14B vs. 8B variance pattern comparison (MMLU).

Component	8B Mean	8B Range	Qwen3-14B	DS-R1-14B
item_id	56.0	51.6–60.2	54.8	50.2
residual	20.7	17.1–22.4	17.2	20.8
temp×item	5.9	4.9–6.9	5.4	8.8
prompt×item	6.2	5.0–7.4	6.0	5.6
seed×item	3.0	2.4–3.6	2.8	5.2
precision	0.3	0.1–0.5	0.2	0.2

Table 17: Fine-tuning variance hierarchy comparison (rank-ordered).

Rank	Inference (MMLU)	Fine-tuning (SST-2)
1	item_id (56.0%)	item_id (40.6%)
2	residual (20.7%)	residual (11.8%)
3	prompt×item (6.2%)	seed×item (10.4%)
4	temp×item (5.9%)	seed (9.2%)
5	seed×item (3.0%)	lr×item (7.8%)
6	prec×item (1.8%)	lr (6.8%)

10 Conclusion

We presented a multi-benchmark G-theory analysis of variance sources in LLM evaluation, crossing infrastructure and methodology facets across five benchmarks, six models, and over 4 million evaluation records. Three findings stand:

1. **Item dominance is benchmark-conditional**: item selection accounts for 21–56% of variance, forming a monotonic 35 pp gradient from knowledge retrieval (MMLU) to complex reasoning (MATH). Chain-of-thought stochasticity systematically shifts variance from item_id into temperature×item and seed×item interactions.

2. **Separability holds at the aggregate**: direct infrastructure×methodology interactions are $<0.3\%$ across all benchmarks, validating isolated-factor studies. Both factor types interact with items, making effects item-conditional.

3. **Optimal protocols reverse across task types**: MMLU needs 420 items (item-limited), GSM8K needs 10 seeds (seed-limited). A "universal reproducibility recipe" is illusory. Current practice ($G < 0.25$) is inadequate on every benchmark.

These findings have immediate practical implications: evaluation protocols should be benchmark-specific, budgets should be allocated based on the dominant variance source for each task type, and the field should adopt reliability reporting alongside accuracy reporting.

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Table 18: Fine-tuning variance across tasks (RoBERTa-large).

Task	N_{train}	item_id	seed	LR	seed×item
SST-2	67K	40.6	9.2	6.8	10.4
MNLI	393K	43.8	7.6	8.4	9.4
RTE	2.5K	35.8	13.2	10.2	11.8
COPA	400	37.4	12.2	9.6	10.8

Table 19: Four-precision mode comparison (MMLU, 4 models pooled).

Mode	Acc.	Agree. w/ FP32	tok/s	VRAM
FP32 (V0 eager)	.664	1.000	1,820	31.2 GB
BF16	.669	.9987	2,760	15.8 GB
FP16	.667	.9984	2,810	15.6 GB
LayerCast	.668	.9993	2,580	16.2 GB

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A MATH Per-Model Breakdown

Table 20: MATH per-model variance breakdown.

Model	item_id	t×i	s×i	Resid.
Qwen3-8B	24.2	19.4	16.2	10.8
Llama-3.1-8B	20.6	21.8	18.4	11.6
Gemma-2-9B-IT	23.4	19.8	16.4	10.4
Mistral-7B-v0.3	17.8	22.8	19.4	12.6

Llama-3.1-8B and Mistral-7B show temperature×item exceeding item_id on MATH, consistent with the expectation that item dominance collapses for lower-accuracy models on extreme reasoning tasks.

B Logprob-Based Scoring

Under logprob-based (continuous) scoring on MMLU, item_id decreases to 42.8% (from 56.0% under binary scoring) because logprobs capture graded confidence rather than binary threshold effects. Prompt main effect rises to 6.8% (from 1.8%), as different prompts systematically shift the logprob distribution. The G coefficient improves to 0.41 (from 0.36), as continuous scoring reduces measurement error.

C Higher-Order Interactions

Three-way interactions total 4.8% of variance, with temperature×seed×item the largest at 1.8%. Extracting three-way terms reduces the residual from 12.0% to 7.2% in the model-as-facet analysis, suggesting that the residual in two-way models is primarily composed of identifiable higher-order interactions rather than truly random noise.